LIMIT THEOREMS

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Limit Theorems

- The most important theoretical results in probability theory are the limit theorems.
- One of these theorems, **"The Central Limit Theorem",** is considered as the most important theorem in probability theory and a foundational concept to inferential statistics.
- Limit theorems will guide us through statistical inference procedures. Therefore it is essential to understand what each theorem states.

Markov's Inequality

- To better understand limit theorems, we must understand the inequalities associated with random variables.
- **Markov's Inequality**[1]**:** If X is a random variable that takes only non-negative values.

  Then, for any positive real value $a > 0$:

$$P\{X \geq a\} \leq \frac{E[X]}{a}$$

[1] Andrey Andreyevich Markov, 1856-1922.

Markov's Inequality: Proof

- **Proof:** Let us assume X is a continuous[2] RV with PDF f. Because $a > 0$, it is sufficient to show

$$aP\{X \geq a\} \leq E[X]$$

to prove Markov's inequality. We can show it as follows:

$$\begin{aligned} aP\{X \geq a\} &= a\int_a^\infty f(x)\,dx \\ &= \int_a^\infty af(x)\,dx \\ &\leq \int_a^\infty xf(x)\,dx \\ &\leq \int_0^\infty xf(x)\,dx \end{aligned}$$

Note that the last integral is equal to $E[X]$, since X is a non-negative random variable. Hence, the proof is complete.

[2] The proof is similar for the discrete case.

Markov's Inequality: Example

Q1. Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50 units. What can be said about the probability that this week's production will be at least 75 units?

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Q1. Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50 units. What can be said about the probability that this week's production will be at least 75 units?

A1. The weekly production quantity, X, is a random variable but its distribution (PDF or CDF) is unknown. We know that X is non-negative, because production quantities can not be negative. We also know that $E[X] = 50$. According to Markov's inequality:

$$P\{X \geq 75\} = P\{X \geq a\} \leq \frac{E[X]}{a} = \frac{50}{75} = \frac{2}{3}$$

We can not compute $P\{X \geq 75\}$ but we can say that it can not be more than $\frac{2}{3}$.

Chebyshev's Inequality

- **Chebyshev's Inequality**[3]**:** Let X be a random variable with mean μ and variance σ^2;

$$E[X] = \mu, \quad Var[X] = \sigma^2$$

Then for any positive real value $k > 0$:

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- Note that, X is not necessarily a normal RV. We have just used μ and σ^2 for a shorter notation.
- Chebyshev's inequality states that the probability of a RV to deviate from its mean more than k units, is less than the ratio of its variance and k^2.

[3] Pafnuty Lvovich Chebyshev, 1821-1894.

Chebyshev's Inequality: Proof

- **Proof:** The probability expression in Chebyshev's inequality has the following equivalence:

$$P\{|X-\mu| \geq k\} = P\left\{(X-\mu)^2 \geq k^2\right\}$$

and according to Markov's inequality:

$$P\left\{(X-\mu)^2 \geq k^2\right\} \leq \frac{E\left[(X-\mu)^2\right]}{k^2} = \frac{Var[X]}{k^2} = \frac{\sigma^2}{k^2}$$

which completes the proof of Chebyshev's inequality.

Chebyshev's Inequality: Example

Q2. Remind yourself with Question 1. Suppose that the variance of the number of items produced during a week is 25. What can be said about the probability that this week's production will be between 40 units and 60 units (boundaries excluded)?

Chebyshev's Inequality: Example (cont'd)

A2. Remember that $E[X] = 50$ The question asks information on:

$$P\{40 < X < 60\} = 1 - P\{|X - 50| \geq 10\}$$

According to Chebyshev's inequality:

$$P\{|X - 50| \geq 10\} = P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2} = \frac{25}{10^2} = 0.25$$

which implies that:

$$P\{40 < X < 60\} \geq 0.75$$

Hence, the probability that this week's production will be between 40 and 60 is at least 0.75.

The Weak Law of Large Numbers

- **The Weak Law of Large Numbers:** Let $X_1, X_2, \ldots$ be a sequence of iid RVs, each having finite mean $E[X_i] = \mu$ and variance $Var[X_i] = \sigma^2$.

  Then, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P\left\{\left|\frac{X_1 + X_2 + \ldots + X_n}{n} - \mu\right| \geq \varepsilon\right\} = 0$$

- The weak law of large numbers states that the probability of the mean of iid RVs to deviate from their expected value more than ε units converges to zero if the number of RVs is large enough, no matter how small ε is chosen.

The Weak Law of Large Numbers: Proof

- **Proof:** To prove the weak law of large numbers, let us compute the expected value and the variance of the mean of these RVs:

$$E\left[\frac{X_1+X_2+\ldots+X_n}{n}\right] = \frac{E\left[\sum_{i=1}^{n} X_i\right]}{n} = \frac{\sum_{i=1}^{n} E[X_i]}{n} = \frac{\sum_{i=1}^{n} \mu}{n} = \frac{n\mu}{n} = \mu$$

and

$$Var\left[\frac{X_1+X_2+\ldots+X_n}{n}\right] = \frac{Var\left[\sum_{i=1}^{n} X_i\right]}{n^2} = \frac{\sum_{i=1}^{n} Var[X_i]}{n^2} = \frac{\sum_{i=1}^{n} \sigma^2}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

- Since X_i are independent, the covariance term is omitted in the previous equation.

The Weak Law of Large Numbers: Proof (cont'd)

- It follows from Chebyshev's inequality that:

$$P\left\{\left|\frac{X_1+X_2+\ldots+X_n}{n}-\mu\right|\geq\varepsilon\right\}\leq\frac{\sigma^2/n}{\varepsilon^2}=\frac{\sigma^2}{n\varepsilon^2}$$

Then, if we take the limit of both sides, we get:

$$\lim_{n\to\infty}P\left\{\left|\frac{X_1+X_2+\ldots+X_n}{n}-\mu\right|\geq\varepsilon\right\}\leq\lim_{n\to\infty}\frac{\sigma^2}{n\varepsilon^2}=0$$

and the weak law of large numbers is proven since a probability statement can be zero at minimum.

$$\lim_{n\to\infty}P\left\{\left|\frac{X_1+X_2+\ldots+X_n}{n}-\mu\right|\geq\varepsilon\right\}=0$$

The Central Limit Theorem

- **The Central Limit Theorem:** Let $X_1, X_2, \ldots$ be a sequence of iid RVs, each having finite mean $E[X_i] = \mu$ and variance $Var[X_i] = \sigma^2$.

  Then the distribution of:

$$\frac{X_1 + X_2 + \ldots + X_n - n\mu}{\sigma\sqrt{n}}$$

  tends to standard normal as $n \to \infty$. In other words, for any real value x:

$$\begin{aligned} P\left\{\frac{X_1+X_2+\ldots+X_n-n\mu}{\sigma\sqrt{n}} \leq x\right\} &= \int_{-\infty}^{x} \phi(t)\,dt \\ &= \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\,dt = \Phi(x) \end{aligned}$$

  if n is sufficiently large.

The Central Limit Theorem (cont'd)

- An immediate result of the central limit theorem is, for a sequence of iid RVs $X_1, X_2, \ldots$, each having mean μ and variance σ^2:

$$X_1 + X_2 + \ldots + X_n \sim N\left(n\mu, \sigma\sqrt{n}\right)$$

  as n goes to infinity.
- In other words, the sum of a sufficiently large number of iid RVs is normally distributed.
- As a consequence, the average of a sufficiently large number of iid RVs is also normally distributed.
- The history and the proof of the central limit theorem can be found in the textbook[4] of Ross (2020).

[4] See pages 219 and 409.

The Central Limit Theorem: Example

Q3. **Normal Approximation to Binomial RV:** There are 336 seats in a theater hall which holds the premiere of a movie. For the premiere 432 tickets are sold, counting on that each guest is likely to miss the premiere with probability 0.25. What is the probability that a guest arrives and can not find an empty seat?

[5] The value is computed as 0.0810 using R.

The Central Limit Theorem: Example

Q3. **Normal Approximation to Binomial RV:** There are 336 seats in a theater hall which holds the premiere of a movie. For the premiere 432 tickets are sold, counting on that each guest is likely to miss the premiere with probability 0.25. What is the probability that a guest arrives and can not find an empty seat?

A3. Let X denote the number of guest who are going to attend:

$$X \sim Binom\,(n = 432, p = 0.75)$$

The desired probability[5] involves computational burden, since it requires computation of complex probabilities:

$$P\{X > 336\} = \sum_{i=337}^{432} \binom{432}{i} (0.75)^i (0.25)^{n-i} = ??$$

[5] The value is computed as 0.0810 using R.

The Central Limit Theorem: Example (cont'd)

A3. Moreover, we can not apply a Poisson approximation since $p = 0.75$ is not a small probability.

- However, the binomial RV here is a sum of a large number of ($n = 432$) iid Bernoulli RVs ($p = 0.75$).
- According to the central limit theorem, if the size parameter, n, of a binomial random variable is large enough (and the success probability, p, is not extreme[6]), it can be approximated with a normal RV with same expected value and variance. In other words:

$$X \sim Binom\left(n = 432, p = 0.75\right), \quad E\left[X\right] = np = 324, \quad Var\left[X\right] = np\left(1-p\right) = 81$$

can be approximated with

$$X' \sim N\left(\mu = 324, \sigma = \sqrt{81} = 9\right)$$

[6] success prob. should not be too close to 0 or 1.

The Central Limit Theorem: Example (cont'd)

A3. Note that, X is an integer valued DRV, whereas X' is a CRV. Hence, for any integer value x, the following approximation is valid:

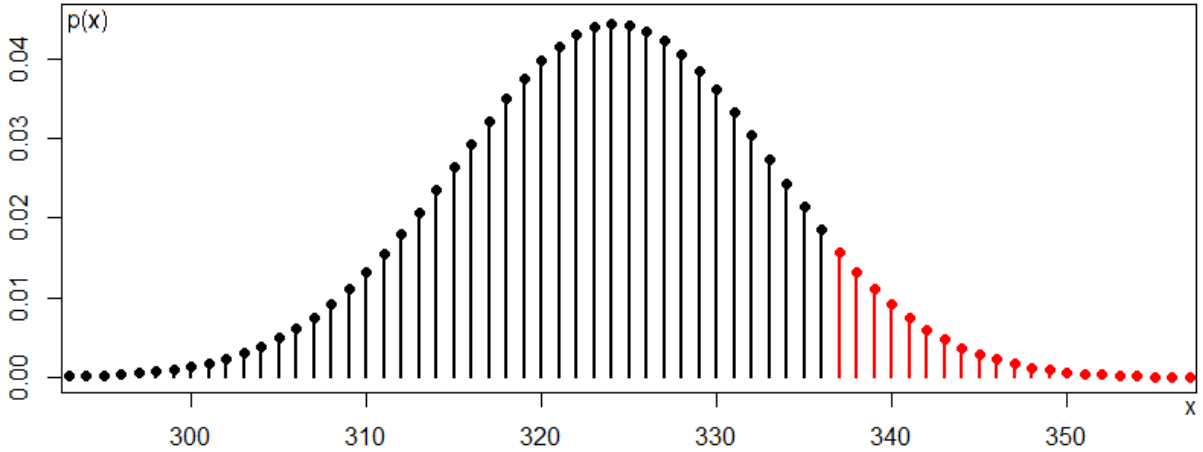
$$P\{X = x\} \approx P\left\{x - 0.5 \leq X' \leq x + 0.5\right\}$$

Hence the desired probability can be approximated as:

$$\begin{aligned} P\{X > 336\} &\approx P\{X' > 336.5\} = P\left\{Z > \tfrac{336.5-324}{9}\right\} \\ &\approx P\{Z > 1.3889\} = 1 - \Phi(1.3889) \\ &\approx 1 - 0.9176 = 0.0824 \end{aligned}$$

- Approximately with 0.0824 probability, a guest may arrive and may not be able to find an empty seat.

The Central Limit Theorem: Example (cont'd)

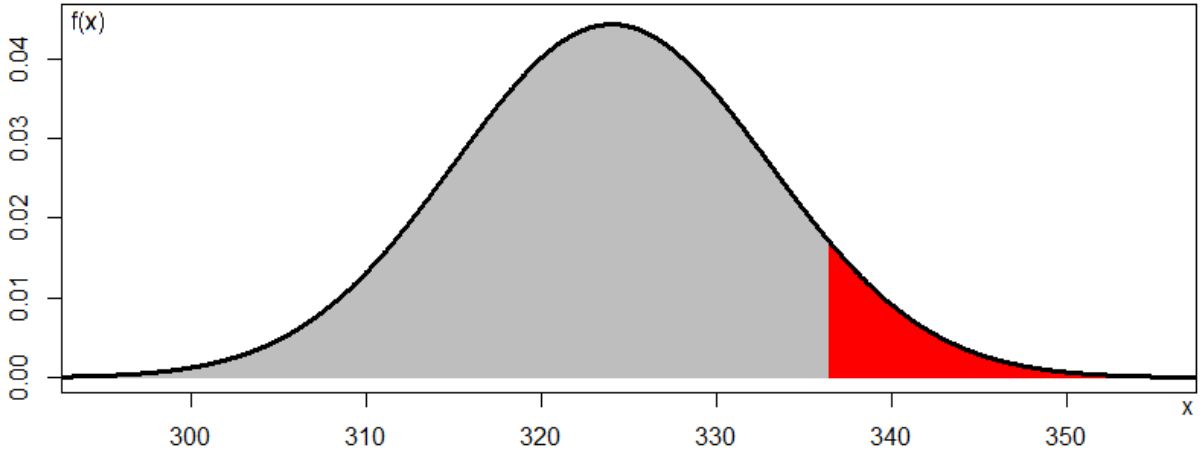


PMF of

$X \sim Binom\left(n = 432, p = \frac{3}{4}\right)$

$\sum_{x=337}^{432} \binom{432}{x} \left(\frac{3}{4}\right)^{x} \left(\frac{1}{4}\right)^{432-x}$

$= 0.0810$



PDF of

$X' \sim N\left(\mu = 324, \sigma = 9\right)$

$1 - \Phi\left(\frac{336.5-324}{9}\right)$

$= 0.0824$

The Central Limit Theorem: Example II

Q4. **Normal Approximation to Poisson RV:** The number of students who enroll in a course is a Poisson RV with mean 100. If 110 or more students enroll, the lecturer will teach the course in two separate sections, whereas if fewer than 110 students enroll, the lecturer will offer a single section. What is the probability that the lecturer will offer two sections?

A4. Let X denote the number of students who will enroll to the course, that is:

$$X \sim Pois\left(\lambda = 100\right)$$

The desired probability[7] involves computational burden, since it requires computation of complex probabilities:

$$P\left\{X \geq 110\right\} = \sum_{i=110}^{\infty} e^{-100} \frac{100^i}{i!} = ??$$

[7] The value is computed as 0.1706 using R.

The Central Limit Theorem: Example II (cont'd)

A4. However, the Poisson RV here has a large rate parameter ($\lambda = 100$). According to the central limit theorem, if the rate parameter, λ, of a Poisson random variable is large enough, it can be approximated with a normal RV with same expected value and variance. In other words:

$$X \sim Pois\left(\lambda = 100\right), \quad E\left[X\right] = \lambda = 100, \quad Var\left[X\right] = \lambda = 100$$

can be approximated with

$$X' \sim N\left(\mu = 100, \sigma = \sqrt{100} = 10\right)$$

The Central Limit Theorem: Example II (cont'd)

A4. Note that, X is an integer valued DRV, whereas X' is a CRV. Hence, for any integer value x, the following approximation is valid:

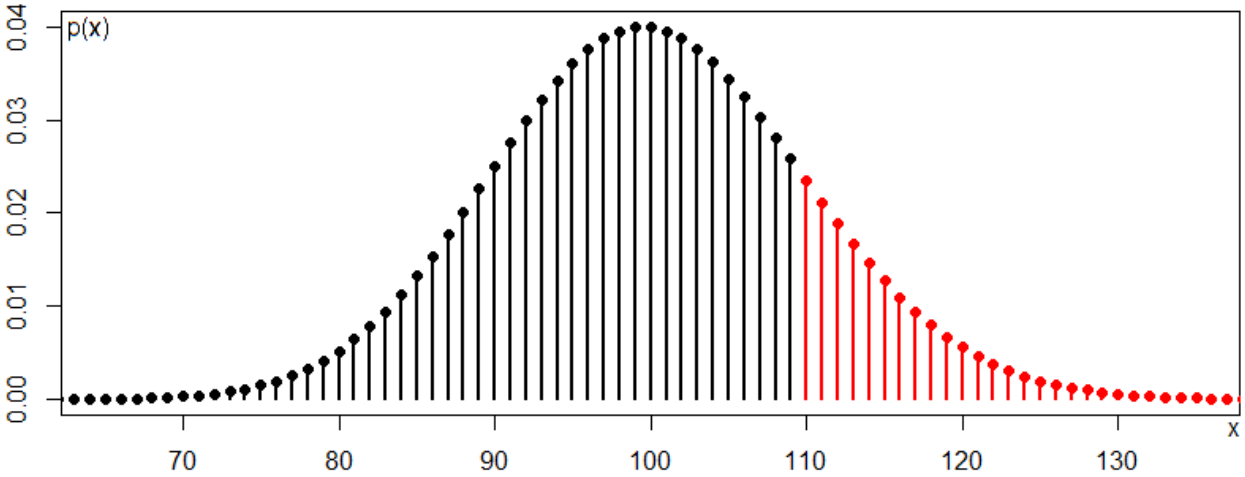
$$P\{X = x\} \approx P\left\{x - 0.5 \leq X' \leq x + 0.5\right\}$$

Hence the desired probability can be approximated as:

$$\begin{aligned} P\{X \geq 110\} &\approx P\{X' > 109.5\} = P\left\{Z > \tfrac{109.5-100}{10}\right\} \\ &\approx P\{Z > 0.95\} = 1 - \Phi(0.95) \\ &\approx 1 - 0.8289 = 0.1711 \end{aligned}$$

- Approximately with 0.1711 probability, the lecturer will offer two sections.

The Central Limit Theorem: Example II (cont'd)

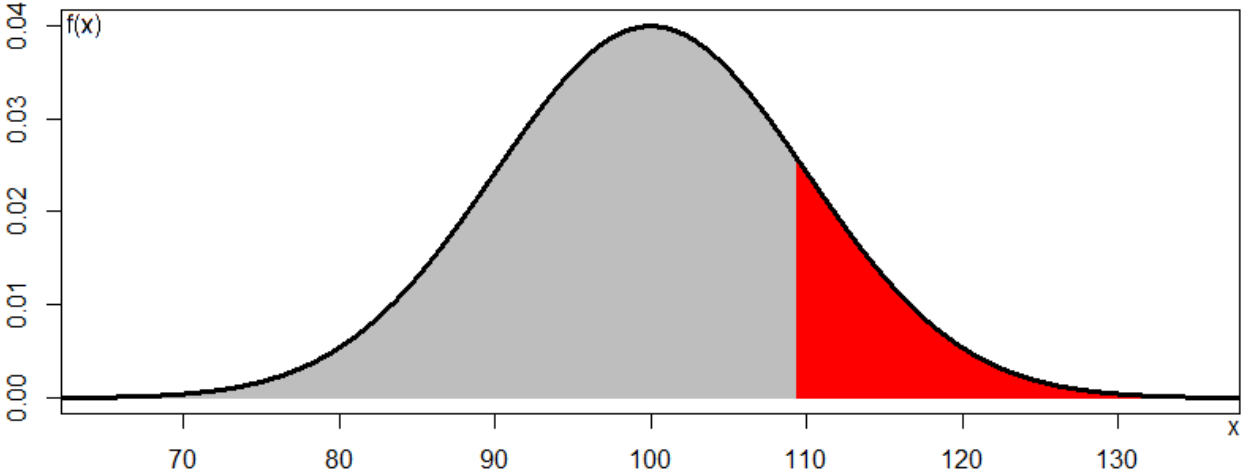


PMF of

$X \sim Pois\,(\lambda = 100)$

$1 - \sum_{x=0}^{109} e^{-100} \frac{100^x}{x!}$

$= 0.1706$



PDF of

$X' \sim N\,(\mu = 100, \sigma = 10)$

$1 - \Phi\left(\frac{109.5-100}{10}\right)$

$= 0.1711$

The Central Limit Theorem: Example III

Q5. **Normal Approximation to Sum of iid RVs:** Let $X_1, \ldots, X_{100}$ be a sequence of iid standard uniform random variables. We define their sum as Y:

$$Y = \sum_{i=1}^{100} X_i$$

Find an approximate solution for $P\{47 < Y < 53\}$.

The Central Limit Theorem: Example III

Q5. **Normal Approximation to Sum of iid RVs:** Let $X_1, \ldots, X_{100}$ be a sequence of iid standard uniform random variables. We define their sum as Y:

$$Y = \sum_{i=1}^{100} X_i$$

Find an approximate solution for $P\{47 < Y < 53\}$.

A5. Because Y is the sum of a sufficiently large number of iid RVs, it can be approximated with normal distribution.

The Central Limit Theorem: Example III (cont'd)

A5. Notice that:

$$X_i \sim Unif\left(\alpha = 0, \beta = 1\right), \quad E\left[X_i\right] = 0.5, \quad Var\left[X_i\right] = \frac{1}{12}$$

The expected value of Y is:

$$E\left[Y\right] = E\left[\sum_{i=1}^{100} X_i\right] = \sum_{i=1}^{100} E\left[X_i\right] = 100 \times 0.5 = 50$$

The variance of Y is:

$$Var\left[Y\right] = Var\left[\sum_{i=1}^{100} X_i\right] = \sum_{i=1}^{100} Var\left[X_i\right] = 100 \times \frac{1}{12} = \frac{100}{12}$$

Note that, the covariance terms are omitted since X_i are independent of each other.

The Central Limit Theorem: Example III (cont'd)

A5. As a result, the following approximation applies:

$$Y \sim N\left(\mu = 50, \sigma = \sqrt{\frac{100}{12}}\right)$$

Since all X_i are continuous RVs, Y is also continuous. So we can directly calculate the desired probability:

$$\begin{aligned} P\{47 < Y < 53\} &= P\left\{\frac{47-50}{\sqrt{100/12}} < Z < \frac{53-50}{\sqrt{100/12}}\right\} \\ &\approx P\{-1.0392 < Z < 1.0392\} \\ &\approx \Phi(1.039) - [1 - \Phi(1.039)] \\ &= 2\Phi(1.039) - 1 \\ &\approx 0.7012 \end{aligned}$$

The Strong Law of Large Numbers

- The strong law of large numbers is the ultimate result of probability theory.
- It is a key to the world of inferential statstics.
- **The Strong Law of Large Numbers**[8]**:** Let $X_1, X_2, \ldots$ be a sequence of iid RVs, each having finite expected value $E[X_i] = \mu$. Then, with probability 1,
$$\frac{X_1 + X_2 + \ldots + X_n}{n} \to \mu$$
as $n \to \infty$.
- The strong law of large numbers states that the average of a sequence of independent RVs having a common distribution will, with probability 1, converge to the mean of that distribution.

[8] See the textbook of Ross (2009) for the proof.

What Is A Random Sample?

- **Parent Population:** A (parent) population is a collection of people, items, or events about which you want to make inferences.
- Because the populations are countably or uncountably large, it is either too expensive or impossible to examine every member of an entire population.
- **Sample:** A finite subset of the population with manageable size.
- The term "sample" may also refer to observed qualitative or quantitative values regarding to the sample.
- **Random Sample:** A (simple) random sample is a sample in which each member of the subset has an equal probability of being chosen.

A Random Sample Example

- **The Parent Population:** Students of Maltepe University.
- **A Simple Random Sample:** The body weights of n randomly selected students.

$$X_1, X_2, \ldots, X_n$$

- The weights $X_1, X_2, \ldots, X_n$ are actually **random variables.** When you randomly re-sample or someone else performs the random sampling, it is unlikely to get the same sample.
- These RVs are **independent** due to the random selection process. The selection of a student does not change the ultimate probability of selecting any other student.
- Moreover, these RVs are also **identical** because they are coming from a common distribution: the same parent population.

A Random Sample Consists of iid RVs

- A **simple random sample** is actually a sequence of independent and identically distributed (iid) random variables, drawn from a larger population.
- In order to make inference about the parent population, this random sample can be analyzed based on our knowledge about probability theory.
  - Suppose we want to learn about the average weight of all students.
  - What can we do if we only know about the weights of the students in our random sample?

Mean of A Random Sample

- Suppose we have a simple random sample $X_1, \ldots, X_n$ of size n coming from a parent population that follows an unknown distribution with unknown mean μ and unknown variance σ^2.
- Let us define the **sample mean:**

$$\bar{X} = \frac{X_1 + \ldots + X_n}{n}$$

- Since the sample is selected randomly[9] **the sample mean is a random variable as well.**

[9] In another selection, we would have possibly selected a different sample.

Expected Value of The Sample Mean

- The expected value of $\bar{X}$ is already shown to be equal to the mean of the parent population.

$$\begin{aligned} E\left[\bar{X}\right] &= E\left[\frac{X_1+\ldots+X_n}{n}\right] \\ &= \frac{E[X_1]+E[X_2]+\ldots+E[X_n]}{n} \\ &= \frac{n\mu}{n} = \mu \end{aligned}$$

- Therefore, the sample mean $\bar{X}$ would be an **unbiased** (accurate) **estimator** (predictor) for the unkown population mean μ.
- According to the strong law of large numbers, the accuracy of prediction increases with a larger sample size.

Variance of The Sample Mean

- Since $\bar{X}$ is a random The variance of the sample mean $\bar{X}$ is already shown to be equal to the variance of the parent population divided by the size of the sample.

$$\begin{aligned} Var[\bar{X}] &= Var\left[\frac{X_1+\ldots+X_n}{n}\right] \\ &= \frac{Var[X_1]+Var[X_2]+\ldots+Var[X_n]}{n^2} \\ &= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \end{aligned}$$

- Though, sample mean $\bar{X}$ is random, its variance can be reduced with a larger sample size. In other words, the variance of the sample mean converges to zero as $n \to \infty$. This is consistent with the strong law.

Other Statistical Properties of The Sample Mean

- Since a random sample is iid, according to the central limit theorem, the sample mean $\bar{X}$ should be normally distributed with a larger sample size. In other words, for large sample sizes:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- Note that, the above assumption does not work for small sample sizes!!!
- How large must the sample size be for a valid normality assumption depends on the distribution of the parent population[10].
- Questionably, most statistics books say that the above assumption is valid when $n \geq 30$.

[10] Watch the video in this link.

Statistical Properties of The Sample Mean: Example

Q6. The daily income of the employed citizens in a town follows an **unknown distribution** with mean 80₺ and variance 100.

a. What is the probability of a randomly selected citizen to have a daily income between 65₺ and 95₺?

b. If we randomly select 200 citizens, what is the probability that their average daily income will be between 78.5₺ and 81.5₺?

c. For the average daily income of 500 random citizens (say $\bar{X}$), provide a half-width x such that:

$$P\{|\bar{X}-80| \leq x\} = P\{80-x \leq \bar{X} \leq 80+x\} = 0.95$$

d. What happens to answer of (c) if we increase the size of our random sample to 2000?

Statistical Properties of The Sample Mean: Ex. (cont'd)

A6. Let X_1 be the daily income of a randomly selected citizen.

a. **It is impossible to calculate this probability since the distribution of the parent population is unknown.** However, using Chebyshev's inequality, we can say that:

$$P\{|X_1 - 80| \geq 15\} \leq \frac{100}{15^2} = \frac{4}{9}$$

Hence:

$$P\{65 \leq X_1 \leq 95\} \geq \frac{5}{9}$$

which states that this probability can not be less than $5/9$. In other words, **at least 5 out of 9 citizens on the average,** have a daily income between 65₺ and 95₺.

Statistical Properties of The Sample Mean: Ex. (cont'd)

b. The parent population has mean $\mu = 80$ and variance $\sigma^2 = 100$. Let us define our random sample as $X_1, \ldots, X_{200}$ and the sample average:

$$\bar{X} = \frac{1}{200} \sum_{i=1}^{200} X_i$$

which is normally distributed according to central limit theorem:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \sim N\left(80, \frac{10}{\sqrt{200}}\right)$$

Hence:

$$\begin{aligned} P\{78.5 \leq \bar{X} \leq 81.5\} &= P\left\{\frac{78.5-80}{10/\sqrt{200}} \leq Z \leq \frac{81.5-80}{10/\sqrt{200}}\right\} \\ &\approx \Phi(2.121) - [1 - \Phi(2.121)] \\ &= 2\Phi(2.121) - 1 \\ &\approx 0.9661 \end{aligned}$$

Statistical Properties of The Sample Mean: Ex. (cont'd)

c. For $n = 500$, sample mean $\bar{X}$ is approximately normal with the following parameterization:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \sim N\left(80, \frac{10}{\sqrt{500}}\right)$$

The following probability is asked to be equal to 0.95:

$$\begin{aligned} P\{80 - x \leq \bar{X} \leq 80 + x\} &= P\left\{\frac{80-x-80}{10/\sqrt{500}} \leq Z \leq \frac{80+x-80}{10/\sqrt{500}}\right\} \\ &= P\{-2.236x \leq Z \leq 2.236x\} \\ &\approx \Phi(2.236x) - [1 - \Phi(2.236x)] \\ &= 2\Phi(2.236x) - 1 = 0.95 \end{aligned}$$

Statistical Properties of The Sample Mean: Ex. (cont'd)

c. We solve the final equation:

$$\begin{aligned} 2\Phi(2.236x) - 1 = 0.95 \quad & \Rightarrow \Phi(2.236x) = 0.975 \\ & \Rightarrow \Phi^{-1}(0.975) \approx 1.96 = 2.236x \\ & \Rightarrow x = \tfrac{1.96}{2.236} \approx 0.877 \end{aligned}$$

Hence, if we sample $n = 500$ random citizens, their daily income average $\bar{X}$ would be between: $80 \pm 0.877 = (79.123, 80.877)$ with 0.95 probability.

Statistical Properties of The Sample Mean: Ex. (cont'd)

d. For $n = 2000$, sample mean $\bar{X}$ is approximately normal with the following parameterization:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \sim N\left(80, \frac{10}{\sqrt{2000}}\right)$$

The following probability is asked to be equal to 0.95:

$$\begin{aligned} P\{80 - x \leq \bar{X} \leq 80 + x\} &= P\left\{\frac{80-x-80}{10/\sqrt{2000}} \leq Z \leq \frac{80+x-80}{10/\sqrt{2000}}\right\} \\ &= P\{-4.472x \leq Z \leq 4.472x\} \\ &\approx \Phi(4.472x) - [1 - \Phi(4.472x)] \\ &= 2\Phi(4.472x) - 1 = 0.95 \end{aligned}$$

Statistical Properties of The Sample Mean: Ex. (cont'd)

d. We solve the final equation:

$$\begin{aligned} 2\Phi(4.472x) - 1 = 0.95 \quad & \Rightarrow \Phi(4.472x) = 0.975 \\ & \Rightarrow \Phi^{-1}(0.975) \approx 1.96 = 4.472x \\ & \Rightarrow x = \tfrac{1.96}{4.472} \approx 0.438 \end{aligned}$$

Hence, if we sample $n = 2000$ random citizens, their daily income average $\bar{X}$ would be between: $80 \pm 0.438 = (79.562, 80.438)$ with 0.95 probability.

- **When we have used a four times larger sample size, the interval is reduced to its half size.**
- **The convergence rate of a sample mean to population mean is proportional to the squareroot of the sample size.**

Variance of A Random Sample

- Suppose we have a simple random sample $X_1, \ldots, X_n$ of size n coming from a parent population that follows an unknown distribution with unknown mean μ and unknown variance σ^2.
- Let us define the **sample variance:**

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}$$

- Since the sample is selected randomly[11] **the sample variance is a random variable as well.**
- But, instead of n, why do we have $n-1$ in the denominator? Let us explain this in the next few slides.

[11] In another selection, we would have possibly selected a different sample.

Expected Value of The Sample Variance

- Let us calculate the expected value of S^2.

$$
\begin{aligned}
E\left[S^2\right] &= \sum_{i=1}^{n} E\left[\frac{1}{n-1}\left(X_i-\bar{X}\right)^2\right] \\
&= \frac{1}{n-1}\sum_{i=1}^{n} E\left[\left(X_i-\bar{X}\right)^2\right] \\
&= \frac{1}{n-1}\sum_{i=1}^{n} E\left[X_i^2-2X_i\bar{X}+\bar{X}^2\right] \\
&= \frac{1}{n-1}\sum_{i=1}^{n} E\left[X_i^2\right]-2E\left[X_i\bar{X}\right]+E\left[\bar{X}^2\right]
\end{aligned}
$$

- In order to continue, we need to calculate the all three expectations within the sum.

Expected Value of The Sample Variance (cont'd)

1. $E\left[X_i^2\right] = Var\left[X_i\right] + E\left[X_i\right]^2 = \sigma^2 + \mu^2$
2. $E\left[\bar{X}^2\right] = Var\left[\bar{X}\right] + E\left[\bar{X}\right]^2 = \frac{\sigma^2}{n} + \mu^2$
3. $$\begin{aligned} E\left[X_i\bar{X}\right] &= E\left[\frac{X_i(X_1+\ldots+X_i+\ldots+X_n)}{n}\right] \\ &= \frac{1}{n}E\left[X_iX_1+\ldots+X_iX_i+\ldots+X_iX_n\right] \\ &= \frac{1}{n}\left(E\left[X_iX_1\right]+E\left[X_iX_i\right]+\ldots+E\left[X_iX_n\right]\right) \end{aligned}$$

Since X_i are independent, $E\left[X_iX_j\right] = E\left[X_i\right]E\left[X_j\right]$ when $i \neq j$.

$$\begin{aligned} E\left[X_i\bar{X}\right] &= \frac{1}{n}\left(E\left[X_i\right]E\left[X_1\right]+\ldots+E\left[X_i^2\right]+\ldots+E\left[X_i\right]E\left[X_n\right]\right) \\ E\left[X_i\bar{X}\right] &= \frac{1}{n}\left(\mu^2+\ldots+\sigma^2+\mu^2+\ldots+\mu^2\right) \\ &= \frac{1}{n}\left(n\mu^2+\sigma^2\right) = \mu^2+\frac{\sigma^2}{n} \end{aligned}$$

Expected Value of The Sample Variance (cont'd)

- Now we can continue calculating the expected value of S^2.

$$
\begin{aligned}
E\left[S^2\right] &= \frac{1}{n-1}\sum_{i=1}^{n} E\left[X_i^2\right] - 2E\left[X_i\bar{X}\right] + E\left[\bar{X}^2\right] \\
&= \frac{1}{n-1}\sum_{i=1}^{n}\left[\sigma^2 + \mu^2 - 2\left(\mu^2 + \frac{\sigma^2}{n}\right) + \frac{\sigma^2}{n} + \mu^2\right] \\
&= \frac{1}{n-1}\sum_{i=1}^{n}\frac{(n-1)\sigma^2}{n} \\
&= \frac{(n-1)\sigma^2}{n-1} = \sigma^2
\end{aligned}
$$

- Since $E\left[S^2\right] = \sigma^2$, the sample variance S^2 would be an unbiased (accurate) estimator (predictor) for the unkown population variance σ^2.

Expected Value of The Sample Variance (cont'd)

- We have emphasized that in the denominator of the sample variance, we have $n-1$.

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}$$

- What would happen if the denominator was n? Then the expectation of the sample variance would be: $E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2\right] = E\left[\frac{n-1}{n}\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2\right] = \frac{n-1}{n}E\left[S^2\right] = \frac{n-1}{n}\sigma^2$ and the sample variance would be a **biased** (inaccurate) estimator of the unknown population variance. Hence, $n-1$ is a correct choice for the denominator.

The Sample Mean and The Sample Variance

- **Summary I:** If we have a parent population following any arbitrary distribution with **unknown mean** μ and **unknown variance** σ^2:
- Draw a sufficiently large sample of size n from the population.
- The sample mean:

$$\bar{X} = \frac{X_1 + \ldots + X_n}{n}$$

would be your best choice to estimate the unknown parent population mean μ.

- The sample variance:

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}$$

would be your best choice to estimate the unknown parent population variance σ^2.

The Sample Variance: Example

Q7. In a production lot of batteries, 8 batteries are randomly selected and exposed to a lifetime test. The following are the observed durabilities of these batteries in hourly time units.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
11.39	8.92	8.83	10.18	9.72	9.41	10.53	8.40

a. Make a guess for the mean lifetime of the batteries in this production lot.

b. Make a guess for the variance of the lifetime of the batteries in this production lot.

c. What should we do to get better guesses of the mean and the variance of the lifetime of the batteries in this production lot.

The Sample Variance: Example

A7. Neither the distribution, nor the mean μ and the variance σ^2 of the lifetime of the batteries in this production lot are known and would be never known unless we measure the lifetimes of each battery in the lot. However, through our random sample of size $n = 8$, we can make guesses for μ and σ^2 values.

a. The sample mean is a guess for the parent population mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{11.39+\ldots+8.40}{8} = 9.9725$$

b. The sample variance is a guess for the parent population variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1} = \frac{(11.39-9.9725)^2+\ldots+(8.40-9.9725)^2}{7} \approx 0.987$$

c. For more accurate estimates (predictions) of μ and σ^2, we should increase our sample size (as long as it is affordable).

Sample Variance When Population Mean Is Known

- Suppose we have a simple random sample $X_1, \ldots, X_n$ of size n coming from a parent population that follows an unknown distribution with **known mean** μ and unknown variance σ^2.
- When the sample variance uses the true population mean, then

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \mu)^2}{n}$$

would be an **unbiased** (accurate) estimator of the unknown population variance. Hence, n would be the correct choice for the denominator.

Sample Variance When Population Mean Is Known (cont'd)

- Let us see, why this is true.

$$
\begin{aligned}
E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2\right] &= E\left[\frac{1}{n}\sum_{i=1}^{n}\left(X_i^2-2\mu X_i+\mu^2\right)\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}E\left[X_i^2-2\mu X_i+\mu^2\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}\left(E\left[X_i^2\right]-2\mu E\left[X_i\right]+E\left[\mu^2\right]\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\left(\sigma^2+\mu^2-2\mu^2+\mu^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\sigma^2=\frac{n\sigma^2}{n}=\sigma^2
\end{aligned}
$$

Sample Variance When Population Mean Is Known (cont'd)

- **Summary II:** If we have a parent population following any arbitrary distribution with **known mean** μ and **unknown variance** σ^2:
- Draw a sufficiently large sample of size n from the population.
- The population mean is known, so the sample mean would not mean anything!
- However, the sample variance formula would change to:

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \mu)^2}{n}$$

  and this time, this would be your best choice to estimate the unknown parent population variance σ^2.
- Instead of the sample mean, when you use the true population mean, you can safely use n in the denominator.

The Sample Var. When Population Mean Is Known: Ex.

Q8. In a production lot of batteries, 8 batteries are randomly selected and exposed to a lifetime test. The following are the observed durabilities of these batteries in hourly time units.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
11.39	8.92	8.83	10.18	9.72	9.41	10.53	8.40

The engineer who designs the batteries says that the batteries are designed to exhaust in 10 hours. In that case, what would be an estimate for the variance of the lifetime of the batteries in this production lot.

The Sample Var. When Pop. Mean Is Known: Ex. (cont'd)

A8. The information that the design engineer provides is an indicator that the mean of the parent population is known and $\mu = 10$. We will use the sample variance formula by replacing:

- sample mean $\bar{x}$ with the true population mean μ,
- the denominator $n-1$ with n.

Therefore, the answer is:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n} = \frac{(11.39-10)^2+\ldots+(8.40-10)^2}{8} \approx 0.971$$

END OF LECTURE SLIDE 11

EPILOGUE

THANKS FOR ALL OF YOUR PARTICIPATION TO MATH 215 CLASS!
WISH YOU AN HEALTHY LIFE AND A BUSINESS CAREER WITH FULL OF FORTUNATE OPPORTUNITIES!
DO NOT MISS THE CHANCES TO HELP PEOPLE AND THE WORLD!
FEEL FREE TO CONTACT WITH ME ANY TIME!
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